Weeks 1 and 2 closeout: corrected seed-42 slides

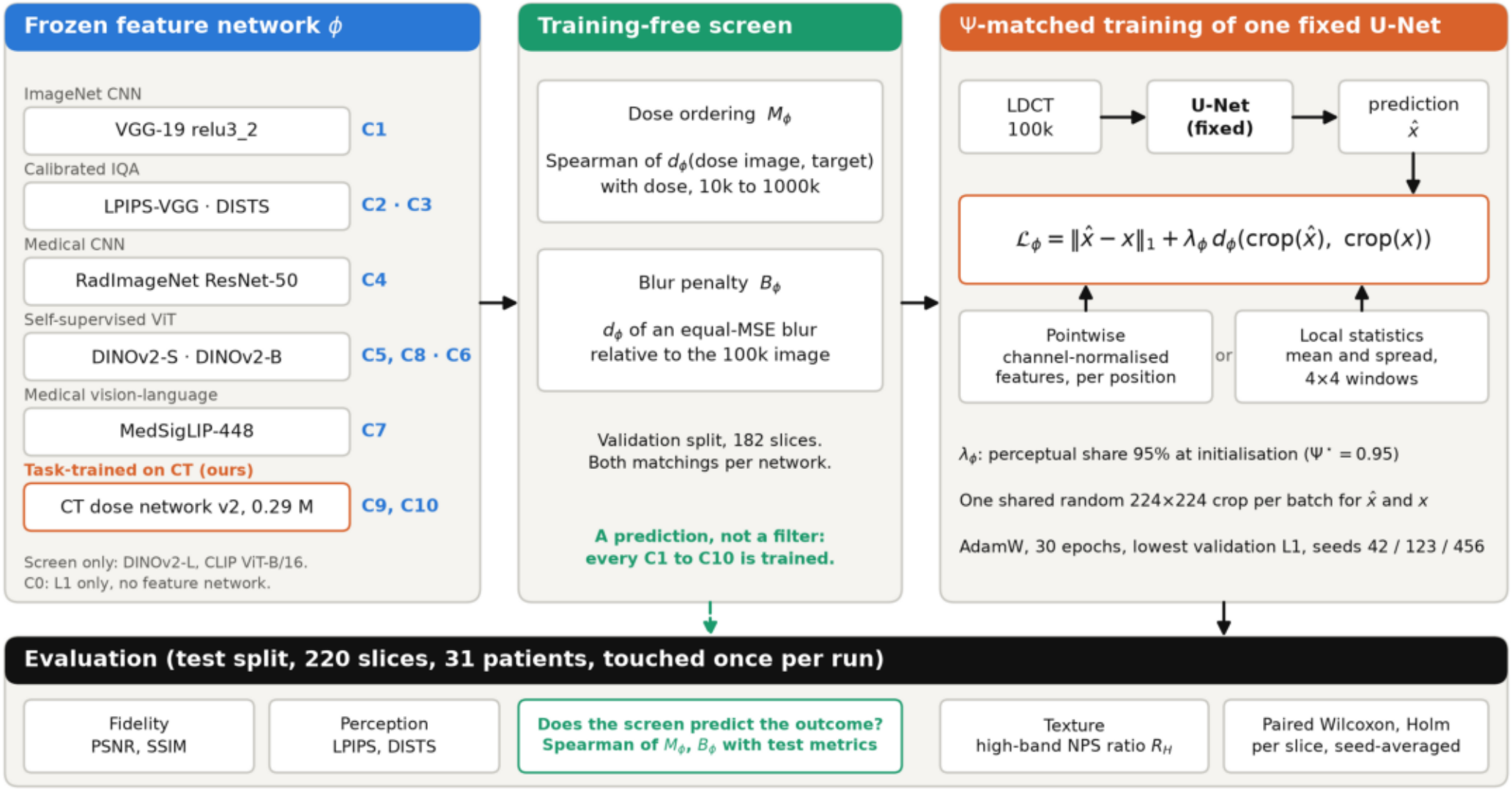
- Replaces the result slides of ldct_backboone_week_1_2.pdf (slides 6 to 12, 14 and 15).
- Final rows: C0 to C7 as trained in Week 2, C8 (DINOv2-S, statistics) after the variance fix, C9 and C10 on the CT dose network v2 (log10-dose regression).
- Superseded runs (pre-fix C8, v1 C9 and C10) are shown once, labelled, for disclosure.
- Every number is read from results/ by build_update_deck.py. One seed; spreads are over test slices.
- Setup, configuration and Week 1 check slides of the original deck are unchanged.

Arda Şahin | BIAI Lab | ISBI 2027 submission

Results freeze Oct 18 · first draft Oct 23 · deadline Oct 26

Overview of the benchmark (Fig 1 draft)

method



Results, seed 42, test split (220 slices)

week 2

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓	DISTS ↓	R_H → 1	ms/step	best epoch
LDCT input (100k)	23.04	0.742	0.284	0.164	6.174	-	-
C0 (L1)	30.88	0.872	0.104	0.112	0.652	118	26
C1 (VGG-19, pointwise)	30.17	0.862	0.060	0.069	0.769	129	27
C2 (LPIPS-VGG, pointwise)	30.73	0.864	0.057	0.063	0.784	141	26
C3 (DISTS, statistics)	30.03	0.804	0.051	0.065	0.836	147	28
C4 (RadImageNet, pointwise)	28.85	0.834	0.083	0.084	0.765	134	27
C5 (DINOv2-S, pointwise)	29.53	0.865	0.060	0.067	0.655	150	23
C6 (DINOv2-B, pointwise)	29.69	0.857	0.064	0.074	0.716	201	19
C7 (MedSigLIP, pointwise)	29.90	0.857	0.070	0.071	0.582	1485	22
C8 (DINOv2-S, statistics), variance fix	**30.55**	**0.867**	**0.049**	**0.060**	**0.860**	**149**	**27**
C9 (CT dose network v2, pointwise)	30.40	0.858	0.068	0.097	0.853	126	26
C10 (CT dose network v2, statistics)	29.71	0.843	0.079	0.102	0.912	129	17

- C8 (DINOv2-S, statistics) is the best trade-off: -0.33 dB against C0 (L1) for the best LPIPS (0.049) and DISTS (0.060).
- C9 and C10 (CT dose network v2) have the highest R_H but stay below C0 on PSNR and SSIM and are the weakest on DISTS.
- One seed. No significance claim; the spread over test slices is in table1_seed42.csv.

Superseded and ablation runs (disclosure only)

week 2

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓	DISTS ↓	R_H → 1	ms/step	best epoch
C8 (DINOv2-S, statistics), before the variance fix	30.53	0.866	0.048	0.060	0.857	153	30
C9 (CT dose network v1, pointwise)	30.26	0.846	0.078	0.100	0.871	128	27
C10 (CT dose network v1, statistics), before the variance fix	30.17	0.854	0.068	0.093	0.875	132	26
C10 (CT dose network v1, statistics), variance fix	29.89	0.837	0.073	0.100	0.879	129	25

- Variance fix: eps inside the square root of the local spread, in place of a clamp that zeroed the gradient in flat windows.
- C8 (DINOv2-S, statistics) is unchanged by the fix (30.53 to 30.55 dB, DISTS 0.060 to 0.060).
- C10 (CT dose network v1, statistics) moved from 30.17 to 29.89 dB with the fix: one seed, of the order of run-to-run variation.
- v1 is the 6-way dose classifier. It is kept as an ablation, never in the final table.

CT dose network v2 and the adoption rule

week 2

Dose network	Matching	ordering M	blur penalty B	dose ratio 10k/1000k
v1 (6-way classifier)	pointwise	0.724	0.308	4.93
v1 (6-way classifier)	statistics	0.712	0.754	2.69
v2 (log10-dose regression)	pointwise	0.971	0.343	4.60
v2 (log10-dose regression)	statistics	0.852	0.863	2.28

- Same three-block feature extractor (287,328 parameters); only the training target changed. Validation screen, 182 slices.
- Ordering M rose for both matchings. B stays below 1: an equal-MSE blur is still cheaper than the 100k noise.
- Disclosure: the adoption threshold was first 0.9 and was lowered to 0.8 after seeing this screen, because statistics matching (0.852) missed 0.9. The rule is not prespecified (results/decision.json).
- The screen gain did not carry over: C9 and C10 on v2 do not improve uniformly on v1 (previous slide).

Planned pairs (descriptive)

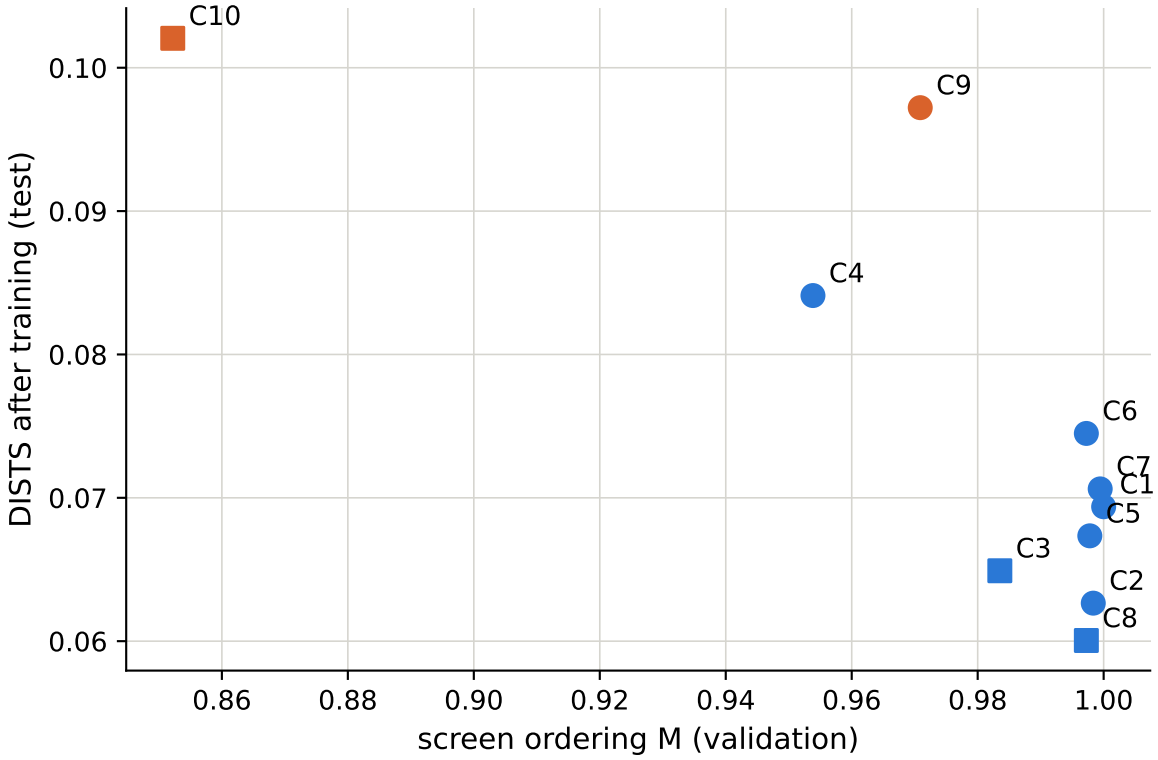
week 2

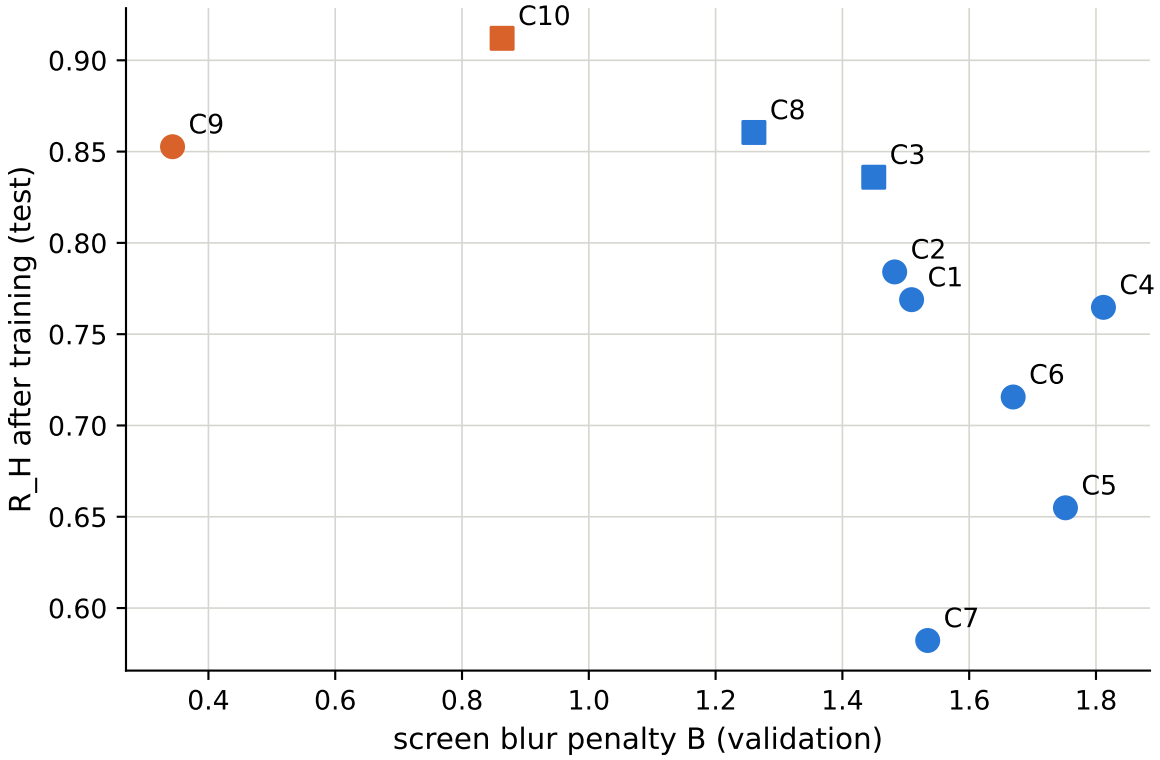
Pair	ΔPSNR	ΔSSIM	ΔLPIPS	ΔDISTS	R_H
C5 → C8 (DINOv2-S, pointwise → statistics)	+1.02	+0.002	-0.011	-0.007	0.655 → 0.860
C9 → C10 (CT dose network v2, pointwise → statistics)	-0.69	-0.015	+0.012	+0.005	0.853 → 0.912
C5 → C6 (pointwise, DINOv2-S → DINOv2-B)	+0.16	-0.008	+0.004	+0.007	0.655 → 0.716

- Matching: statistics improve DINOv2-S on every metric.
- CT dose network v2: statistics raise R_H but cost PSNR, SSIM, LPIPS and DISTS. The sign differs from DINOv2-S.
- Size: DINOv2-B shows no benefit over DINOv2-S and is slower.
- No significance claim. Wilcoxon tests with Holm correction follow with three seeds.

Screen versus trained outcome

preliminary

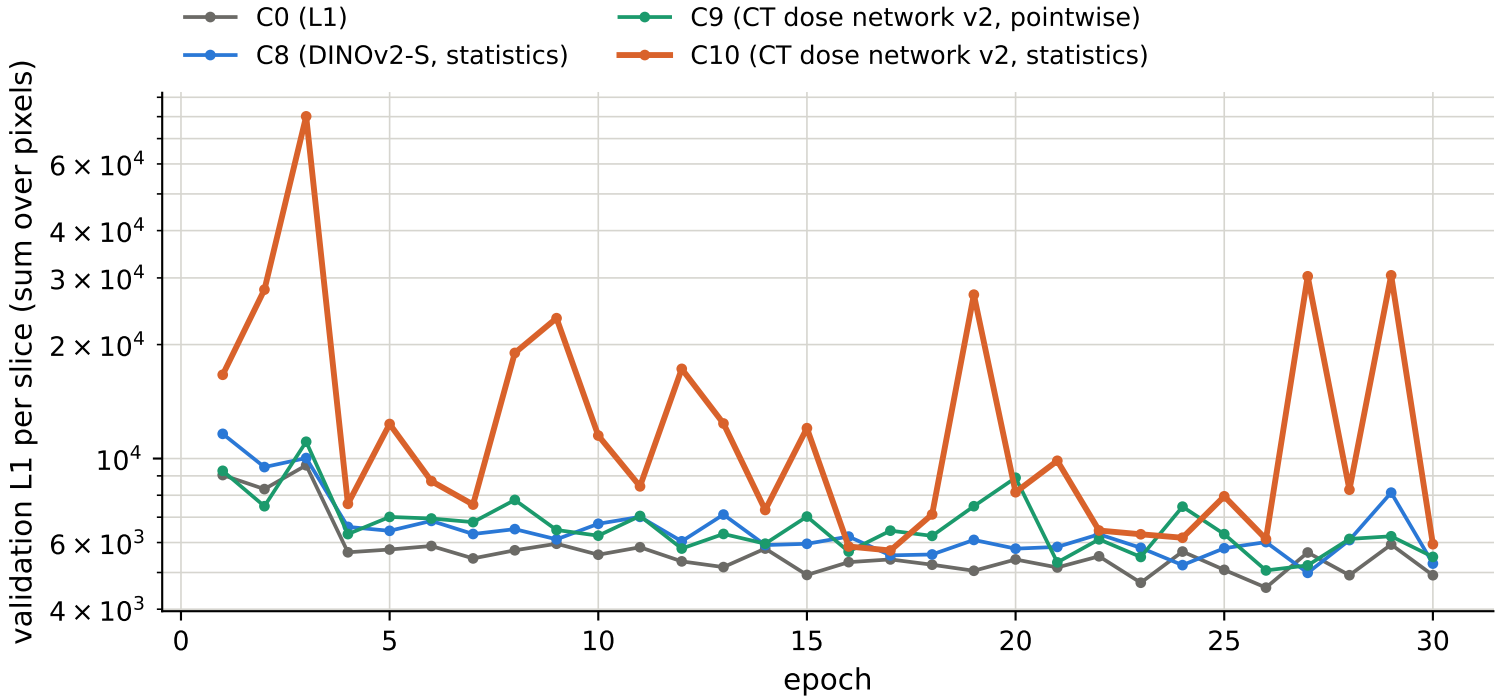


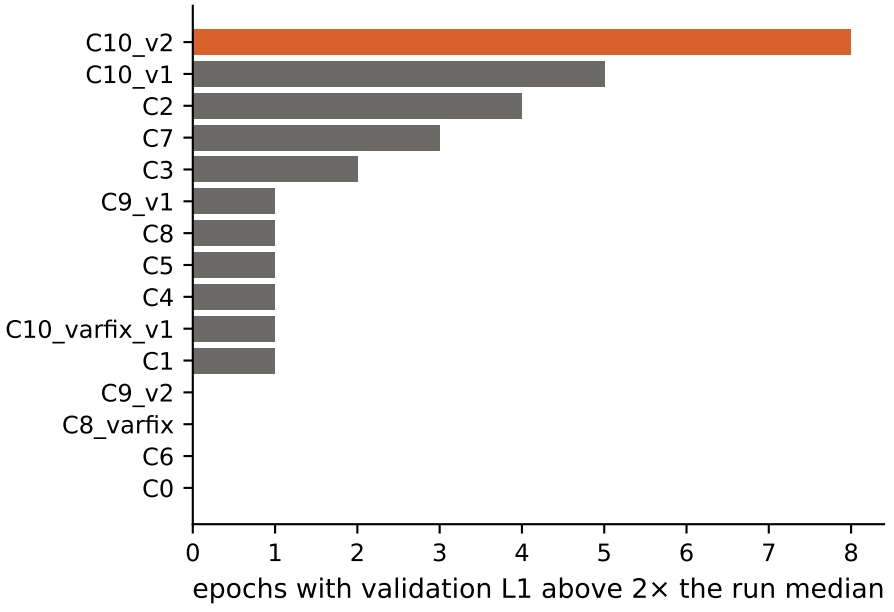


- Circle: pointwise. Square: statistics. Orange: CT dose network v2. Ten trained networks, one seed.
- v2 raised M to 0.97 and 0.85, yet C9 and C10 keep the worst DISTS: M alone does not predict the outcome.
- B below 1 predicted smoothing; C9 and C10 have the highest R_H instead. The screen lacks a structure term.
- The Spearman tests of screen against outcome are a Week 3 deliverable (three seeds).

C10 validation instability

diagnostics





- C10 (CT dose network v2, statistics): 8 of 30 epochs above twice the median validation L1 (largest 80,138 at epoch 3, 30,478 at epoch 29).
- Training loss falls smoothly; 0 non-finite steps. Spikes also occur with LPIPS-VGG, MedSigLIP, RadImageNet and DISTS, never with L1.
- Checked on CPU: the saved checkpoint reproduces the logged best validation L1 (5,717 against 5,716); epoch 17 is the right choice.
- At its best epoch C10 looks like C0 (L1): no intensity offset, no failing slices, same sensitivity to the BatchNorm mode.
- No bug found. The cause of the spikes is open: the GPU replay with per-slice diagnostics has not been run yet.

Table 2 and compute

resources

Config	Network	Matching	Parameters	λ	ms/step	M	B
C1	VGG-19 relu3_2	pointwise	1.15 M	4.699	129	1.000	1.509
C2	LPIPS-VGG	pointwise	14.71 M	9.162	141	0.998	1.482
C3	DISTS	statistics	14.71 M	6.984	147	0.984	1.449
C4	RadImageNet ResNet-50	pointwise	25.56 M	2.211	134	0.954	1.812
C5	DINOv2-S	pointwise	22.06 M	1.634	150	0.998	1.752
C6	DINOv2-B	pointwise	86.58 M	2.497	201	0.997	1.669
C7	MedSigLIP-448	pointwise	428.57 M	3.755	1485	0.999	1.535
C8	DINOv2-S	statistics	22.06 M	1.966	149	0.997	1.260
C9	CT dose network v2 (regression)	pointwise	0.29 M	1.549	126	0.971	0.343
C10	CT dose network v2 (regression)	statistics	0.29 M	2.802	129	0.852	0.863

- CT dose network: 287,328 parameters are the three blocks the loss reads; the whole network with its head has 1,173,857.
- λ and ms/step are those of the final runs. The Step 1.3 distance check was not rerun for the v2 dose network.
- GPU hours: final seed-42 runs 3.242, superseded runs 0.693, three-seed protocol about 10.990 (A100 80 GB).

Takeaways

- Perceptual terms lower LPIPS by 20 to 53% and DISTS by 8 to 46% against C0 (L1), at a cost of 0.15 to 2.0 dB.
- C8 (DINOv2-S, statistics) is the practical candidate: best LPIPS and DISTS, 0.33 dB below C0.
- Statistics matching helps DINOv2-S on every metric. For the CT dose network v2 it raises R_H only.
- CT dose network v2 orders dose levels much better than v1, but C9 and C10 did not improve uniformly:

  the screen's M does not predict the trained outcome on its own. This is the scientific question for Week 3.
- Reproducibility: per-run source revision and checkpoint hashes are in results/provenance.json;

  seeds 123 and 456 run from seeds_perceptual_backbone_ldct.ipynb with frozen λ and the frozen v2 network.
- Status: one seed. Statistical tests, three-seed tables and final figures belong to the Oct 18 freeze.